

Operating Systems - Process Management

Introduction

An Operating System (OS) is system software that manages computer hardware and software resources. It acts as an interface between the user and the computer hardware. The operating system provides services such as process management, memory management, file management, and device management.

Process

A process is a program that is currently being executed. Every process has its own Process Control Block (PCB), which stores information such as the process ID, process state, program counter, CPU registers, scheduling information, and memory allocation details.

Process States

A process generally passes through the following five states:

- New
- Ready
- Running
- Waiting (Blocked)
- Terminated

The operating system changes the process from one state to another depending on CPU scheduling and resource availability.

CPU Scheduling

CPU scheduling is the process of selecting which process should execute next.

Common scheduling algorithms include:

- First Come First Serve (FCFS)
- Shortest Job First (SJF)
- Round Robin (RR)
- Priority Scheduling

